Inference-Time Refinement Closes the Synthetic-Real Gap in Tabular Diffusion

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Abstract

Diffusion-based generators set the current state of the art for synthetic tabular data, deployed downstream wherever direct access to real records is restricted. These methods approach but rarely exceed real-data utility on downstream tasks, and closing this synthetic–real performance gap has so far been pursued mostly at training time, via architectural advances, scaling, and retraining of monolithic generators. The inference-time alternative, i.e., refining the outputs of a pre-trained backbone with parameters left untouched, has remained largely unexplored for tabular synthesis. We introduce **TARDIS** (Tabular generation through Refinement, Distillation, and Inference-time Sampling), an inference-time search framework over a frozen pre-trained backbone: per dataset, a Tree-structured Parzen Estimator configures score-level guidance during reverse diffusion, with each trial's objective set by an inner grid search over post-hoc sample selectors and an optional soft-label distillation step. The search space encodes a mathematical pattern we name *Bidirectional Chamfer Refinement* (BCR): the symmetric Chamfer functional between synthetic and real samples is minimized both continuously, via a score-level gradient during reverse diffusion, and discretely, via batch-ranking post-generation. On 9 of 15 datasets the search selects BCR-aligned configurations over the alternatives encoded in the search space. Across 15 binary, multiclass, and regression benchmarks and at the cardinality selected by the search, TARDIS achieves a median $+8.6\%$ downstream-task improvement over models trained on real data (mean $+9.8\%$, 95% CI of the mean $[+3.8, +16.4]$, Wilcoxon $p = 0.008$, 12/15 strict wins) and improves over the underlying TabDiff backbone on all 15 datasets (mean $+13.1\%$, $p < 10^{-4}$), matching the backbone on manifold fidelity, diversity, and sample-level privacy proxies. At output size matched to the real data, it still improves over the backbone on 11 of 15 datasets while reaching parity with real data: cardinality selection is an integral part of the method. The synthetic–real gap is therefore not primarily a training-time problem: on the studied corpus, inference-time refinement of a pre-trained tabular diffusion backbone reaches and exceeds real-data utility, at an apply-time cost of 1 to 80 minutes on a single consumer-grade GPU after a one-time configuration search. Code: `https://github.com/[TODO]`

1 Introduction

Mixed-type tabular records remain the dominant input format of applied machine learning. When direct access to such records is restricted, synthetic generation enables downstream training [13, 12], and diffusion-based generators currently set the state of the art [35]. Despite steady architectural progress, downstream learners trained on the output of these generators approach but rarely match the utility of equivalent learners trained on real data, leaving a persistent *synthetic–real gap*.

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Most existing approaches to closing this gap follow a single paradigm: improve the generator. Each new architecture, e.g., TabDiff [31] with its per-feature noise schedules, adds complexity but yields only marginal gains on the headline downstream metric, leaving real-data utility unmatched on most standard benchmarks. The implicit assumption underlying this trajectory is that the gap must be closed at training time, with the generator as the only target of intervention.

An alternative paradigm has emerged in image synthesis, where inference-time guidance methods modify the sampling process of a pre-trained diffusion model at no additional training cost. Classifier-free guidance [18] steers sampling toward class-conditional or prompt-conditional outputs by interpolating score predictions; Chamfer Guidance [11] steers it toward arbitrary user-specified reference batches by perturbing the score with the gradient of a bidirectional Chamfer functional. Direct transfer of this paradigm to tabular synthesis is non-trivial, however, due to (i) mixed-type feature spaces that lack a canonical differentiable distance, (ii) the absence of a standardized representation space analogous to CLIP for images, and (iii) the dual fidelity-utility objective specific to the train-on-synthetic-test-on-real (TSTR) setting.

At its core, our approach generates many candidate samples from the pre-trained backbone and retains those that best match the real-data manifold, both by steering generation toward it and by filtering the output pool afterward. We call this framework **TARDIS** (Tabular generation through Refinement, Distillation, and Inference-time Sampling), an inference-time refinement framework that operates on a frozen pre-trained tabular diffusion backbone. TARDIS is grounded in the *Bidirectional Chamfer Refinement* (BCR) principle: a continuous score-level perturbation during reverse diffusion and a discrete post-generation subsampling step both minimize the same symmetric Chamfer functional between candidates and a real reference batch, with the continuous step concentrating probability mass near the real-data manifold and the discrete step pruning residual off-manifold candidates. BCR addresses the three obstacles above in unified form: it handles (i) and (ii) through a representation map φ instantiated per dataset either as the identity on z-score-normalized features or as the encoder of a tabular β-VAE, providing a differentiable representation that absorbs mixed-type heterogeneity without requiring a canonical metric; and (iii) by balancing fidelity (synthetic-to-real) and coverage (real-to-synthetic) terms in a single bidirectional functional, coupling the TSTR utility objective directly to manifold-distance minimization.

The framework operationalizes BCR through three stages (Figure 1): Stage I oversamples a noise pool, Stage II applies the score-level Chamfer perturbation during reverse diffusion, and Stage III ranks the resulting candidate pool under the same functional, optionally after a soft-label distillation step [17, 26]. Stage II hyperparameters are tuned per dataset by a Tree-structured Parzen Estimator search [4]; at each trial Stage III runs a grid over selectors and the distillation toggle, returning its best validation utility as the trial's objective. TARDIS operates in the standard synthetic-data-release setting: the generator is trained on the real records and only synthetic data are released; guidance, selection, distillation, and configuration search use the real records in the same environment as backbone training, requiring no access beyond what training the backbone itself already requires.

Empirically, across 15 heterogeneous tabular benchmarks (binary classification, multiclass classification, regression), TARDIS deployed at its selected cardinality surpasses real-data utility on 12 of 15 datasets at a median $+\mathbf{8.6}\%$ improvement on the headline downstream metric, improves over the underlying TabDiff backbone on all 15, and matches the backbone on manifold fidelity, diversity, and sample-level privacy proxies; at matched output size ($K = 1$) it still improves over the backbone on 11 of 15 datasets while reaching parity with real data. Cardinality selection is thus an integral part of the method. On this corpus, the synthetic–real gap is therefore not primarily a training-time problem.

TARDIS is, to the best of our knowledge, the first inference-time search framework for tabular diffusion; BCR is the unifying principle coupling its continuous and discrete steps under a single Chamfer functional, and the region of the search space elected most often; and our empirical results show that inference-time refinement, including cardinality selection, suffices to close the synthetic–real gap on the studied corpus without retraining.

2 Related Work

We review prior work along three axes that intersect in TARDIS, i.e., tabular generative modelling, inference-time guidance for diffusion, and post-hoc curation of generated samples.

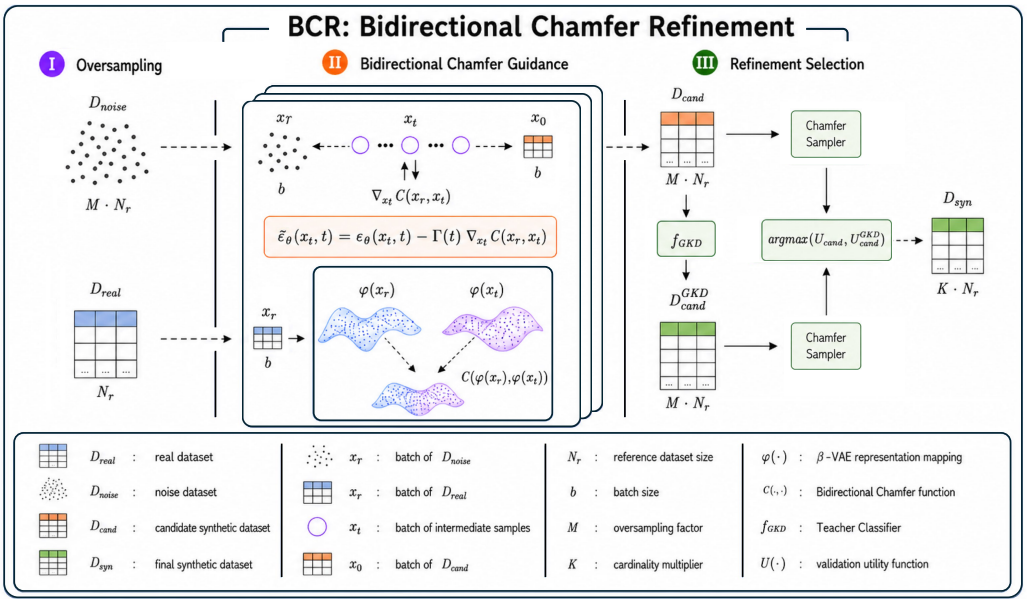


Figure 1: **TARDIS pipeline.** Stage I draws an oversampled noise pool D_{noise} of cardinality $M \cdot N_r$ from the latent space $\mathcal{Z}$. Stage II denoises D_{noise} via reverse diffusion, perturbing the score $\varepsilon_\theta(x_t, t)$ with the gradient of the bidirectional Chamfer functional $\mathcal{C}$ between the current candidate batch x_t and a real reference batch x_r, projected through a representation map φ; this produces the candidate pool D_{cand}. Stage III selects D_{syn} of cardinality $K \cdot N_r$ from D_{cand} via a Chamfer Sampler, optionally preceded by Generative Knowledge Distillation f_{GKD} that relabels candidates with soft targets; the final variant is chosen by argmax over the validation utility $\mathcal{U}$. The figure illustrates the BCR-headline configuration with β-VAE mapping and Chamfer Sampler. The full TARDIS framework generalizes Stage II to also support disabled-guidance and identity-mapping variants, and Stage III to a grid over five samplers (Section 3.3, Appendix D).

Tabular generative models. Existing approaches treat the synthetic–real utility gap as a problem of the generator architecture. GAN-based methods, e.g., CTGAN [40] and CTAB-GAN+ [44], address mixed-type generation through conditional sampling and mode-specific normalization; VAE and autoregressive families, with TVAE [40] and TabularARGN [33], factorize the joint via variational decoders or per-feature autoregressive heads; diffusion-based formulations, with TabDDPM [22], STaSy [20], CoDi [24], TabSyn [43], and TabDiff [31], set the current state of the art through increasingly elaborate score-based and latent-space designs. Each step advances the architecture, but none operates on the output distribution of an already-trained backbone. TARDIS departs from this trajectory by refining that output distribution at inference time, with the backbone left frozen.

Inference-time guidance for diffusion. Inference-time guidance steers a pre-trained diffusion model toward task-specific or user-specified outputs without retraining. Building on standard score-based diffusion [19], classifier-free guidance [18] interpolates between conditional and unconditional score predictions for class- and prompt-conditional image synthesis; Chamfer Guidance [11] extends this to reference-based steering by perturbing the score with the gradient of a bidirectional Chamfer functional between candidates and a reference set in image space. Tabular counterparts are absent in the literature. TARDIS adapts the Chamfer construction to the tabular setting through a representation map φ instantiated per dataset (identity on z-score-normalized features or the encoder of a β-VAE), and adds a discrete post-hoc selection stage that Chamfer Guidance does not include, unifying both levels under the same functional.

Synthetic-data filtering and post-hoc refinement. Post-hoc curation of generated samples emerged in image synthesis with the Chamfer Sampler of [11], which ranks an oversampled candidate pool by bidirectional manifold proximity to a real reference set. In tabular synthesis the closest precedent is ISSOSYNTH [16], which oversamples a pre-trained generator and selects a subset minimizing a binned L_1 distance over low-order marginals via discrete iterative-proportional-fitting. Soft-label distillation has been used in tabular and image synthesis to inject teacher knowledge into synthetic

training sets [17, 26]; feature-space oversampling for class imbalance, e.g., SMOTE [8], operates without a learned metric or generative pool. TARDIS reframes the oversample-and-select pattern of ISSOSYNTH as one of two coupled operational levels, pairing the discrete selection step with continuous score-level guidance during reverse diffusion, replacing binned-marginal distances with a continuous metric in φ-space, and unifying both levels under a single Chamfer functional.

Across these three clusters, no prior work occupies the intersection TARDIS targets: *post-pretraining* refinement of a tabular diffusion backbone, with continuous and discrete steps unified under a single Chamfer functional. The closest precedents are Chamfer Guidance [11] in the image domain (continuous guidance only) and ISSOSYNTH [16] in tabular synthesis (discrete selection only); TARDIS combines and extends both.

3 Methodology

We introduce **TARDIS** (Tabular generation through Refinement, Distillation, and Inference-time Sampling), a three-stage inference-time refinement framework operating on a frozen pre-trained tabular diffusion backbone $G_\theta : \mathcal{Z} \to \mathcal{X} \times \mathcal{Y}$, where $\mathcal{Z}$ is the latent noise space, $\mathcal{X} = \mathcal{X}_{\text{num}} \times \mathcal{X}_{\text{cat}}$ the mixed-type feature space, and $\mathcal{Y}$ the target space ($\mathbb{R}$ for regression, finite for classification). Given a real dataset $D_{\text{real}} \subset \mathcal{X} \times \mathcal{Y}$ of cardinality N_r and a held-out test set D_{test}, TARDIS produces a synthetic dataset D_{syn} of cardinality $K \cdot N_r$ targeting the train-on-synthetic-test-on-real (TSTR) utility

$$\mathcal{U}_{\text{syn}} = \psi\big(A(D_{\text{syn}}), D_{\text{test}}\big), \qquad \mathcal{U}_{\text{real}} = \psi\big(A(D_{\text{real}}), D_{\text{test}}\big), \tag{1}$$

where A is a downstream learner and ψ a task-appropriate metric (AUROC, RMSE). The synthetic–real gap is the empirical regularity $\mathcal{U}_{\text{syn}} < \mathcal{U}_{\text{real}}$ across mainstream tabular generators; TARDIS aims to close, and where possible reverse, this gap at inference time.

The framework comprises three stages (Figure 1). Stage I draws an oversampled noise pool $D_{\text{noise}} \subset \mathcal{Z}$ of cardinality $M \cdot N_r$. Stage II denoises D_{noise} through the reverse-diffusion process of G_θ, optionally perturbing the predicted score at each step with the gradient of a bidirectional Chamfer functional against D_{real}, yielding the candidate pool D_{cand} of cardinality $M \cdot N_r$. Stage III selects D_{syn} from D_{cand} via post-hoc batch ranking under the same functional, optionally preceded by a soft-label distillation step. Stages II and III thus minimize the same Chamfer functional $\mathcal{C}$ at two complementary operational levels: continuously, via score-level gradient descent during sampling, and discretely, via batch ranking after sampling. We refer to this joint minimization as *Bidirectional Chamfer Refinement* (BCR). Stage II hyperparameters are tuned per dataset by a Tree-structured Parzen Estimator (TPE) search [4]; at each trial, Stage III performs an inner grid search over its selector and distillation toggle, and the maximum validation utility across that grid is returned as the trial's objective.

Notation. $N_r = |D_{\text{real}}|$ denotes the number of real training rows, M the oversampling factor (a pool of $M \cdot N_r$ candidates; $M = 50$ in all reported configurations), K the cardinality multiplier ($K \cdot N_r$ synthetic rows; $K \in [0.1, 25]$, searched per dataset), φ the representation map in which $\mathcal{C}$ is computed, and $\Gamma(t)$ the Stage II guidance scale.

3.1 Stage I: Oversampling

Stage I draws $M \cdot N_r$ samples from the latent prior of $\mathcal{Z}$, forming the noise pool D_{noise}. The oversampling factor M controls the redundancy available to Stage III: each retained sample is selected from M/K candidates. We fix $M = 50$ across all benchmarks. BCR requires $M > K$ so that the Stage III refinement operates on a non-trivial selection set; the saturation of utility as $K \to M$ is characterized in Appendix A.

3.2 Stage II: Bidirectional Chamfer Guidance

Stage II denoises D_{noise} through the reverse-diffusion process of G_θ. During reverse diffusion the predicted score is optionally perturbed by the gradient of a bidirectional Chamfer functional against D_{real}, steering candidates toward the real-data manifold. The resulting candidate pool D_{cand} has cardinality $M \cdot N_r$. The TPE search introduced above selects all Stage II hyperparameters jointly per dataset; the full search space is reported in Appendix F, Table 7.

Bidirectional Chamfer functional. Let $\varphi : \mathcal{X} \to \mathbb{R}^d$, with $d \in \mathbb{N}^+$, be a representation map and $A, B \subset \mathcal{X}$ two finite sample sets. We adopt the symmetric Chamfer functional [11],

$$\mathcal{C}(A,B) \;=\; \underbrace{\frac{1}{|A|}\sum_{a\in A}\min_{b\in B}\big\|\varphi(a)-\varphi(b)\big\|_2}_{\textit{fidelity}} \;+\; \underbrace{\frac{1}{|B|}\sum_{b\in B}\min_{a\in A}\big\|\varphi(a)-\varphi(b)\big\|_2}_{\textit{coverage}}, \tag{2}$$

which decomposes into a fidelity term penalizing deviations from A to its nearest match in B and a coverage term penalizing the symmetric direction. $\mathcal{C}$ is symmetric, non-negative, and zero if and only if $\varphi(A) = \varphi(B)$ as multisets (Appendix A).

Guidance variants. The TPE search exposes guidance as a binary toggle. When disabled, the score perturbation of Equation 4 is dropped and Stage II reduces to standard reverse diffusion. When enabled, the representation map φ admits two mutually exclusive forms, also exposed to the TPE search: (i) *identity mapping*, with φ the identity on z-score-normalized features and the Chamfer gradient computed in the data space; (ii) β*-VAE mapping*, with φ the encoder of a tabular β-VAE [21, 6] learned per dataset on D_{real} and the Chamfer gradient computed in its latent space. Figure 1 illustrates the β-VAE-mapping variant.

β-VAE encoder. The encoder is trained on D_{real} under the standard β-VAE evidence lower bound,

$$\mathcal{L}_{\text{VAE}}(x) \;=\; \mathcal{L}_{\text{rec}}(x,\hat{x}) \;+\; \beta(t)\cdot D_{\text{KL}}\big(q_\varphi(z \mid x)\,\|\,p(z)\big)\,, \tag{3}$$

with prior $p(z) = \mathcal{N}(0, I)$, KL annealing $\beta(t)$ from one of five schedule families, and a composite reconstruction loss combining mean squared error on numerical features with cross-entropy on categorical groups, averaged within group to balance the gradient contribution across feature types. The latent dimension $d \in \{4, 8, 16, 32\}$ is selected by the TPE search; full schedule families and training hyperparameters are deferred to Appendix C.

Score perturbation. At each guidance step t, the perturbed score is

$$\tilde{\varepsilon}_\theta(x_t,t) \;=\; \varepsilon_\theta(x_t,t) \;-\; \Gamma(t)\cdot\nabla_{x_t}\mathcal{C}\big(x_t,x_r\big)\,, \tag{4}$$

where ε_θ is the score predicted by the backbone, $\Gamma(t) \geq 0$ a time-dependent scaling factor, and $x_r \subset D_{\text{real}}$ a reference batch sampled at the start of each generation and held fixed across diffusion steps. Guidance is applied at t_g steps of the reverse trajectory rather than at every step, with t_g tuned per dataset (Table 7); the remaining steps use the unperturbed score ε_θ. Equation 4 is an approximation: unlike classifier-free guidance [18], it does not correspond to exact posterior sampling under a likelihood defined by $\mathcal{C}$, and $\Gamma(t)$ keeps the perturbation small relative to ε_θ, acting as a soft steering signal. The schedule $\Gamma(t)$ is selected from four families (constant, linear, cosine, sine) bounded by $\gamma_{\min}, \gamma_{\max}$; the reference batch x_r is either class-conditional or global.

3.3 Stage III: Refinement Selection

Stage III selects D_{syn} of cardinality $K \cdot N_r$ from the candidate pool D_{cand} produced by Stage II. The selection is performed by an inner grid search over a binary Generative Knowledge Distillation (GKD) toggle and five sampler variants (10 combinations); for each TPE trial of Stage II, the configuration maximizing validation utility is retained, and the resulting maximum is returned to the outer search as that trial's objective.

Chamfer Sampler. The Chamfer Sampler is the discrete counterpart of the Stage II score perturbation: it minimizes the same functional $\mathcal{C}$ of Equation 2, evaluated at the pair (B_k, D_{real}), by post-hoc batch ranking rather than gradient descent. Partitioning D_{cand} into batches $\{B_k\}_{k=1}^{M\cdot N_r/b}$ of size b,

$$\mathcal{C}(B_k, D_{\text{real}}) \;=\; \underbrace{\frac{1}{|B_k|}\sum_{x\in B_k}\min_{r\in D_{\text{real}}}\|\varphi(x)-\varphi(r)\|_2}_{m_{\text{sr}}(B_k)\text{: fidelity (synthetic-to-real)}} \;+\; \underbrace{\frac{1}{|D_{\text{real}}|}\sum_{r\in D_{\text{real}}}\min_{x\in B_k}\|\varphi(x)-\varphi(r)\|_2}_{m_{\text{rs}}(B_k)\text{: coverage (real-to-synthetic)}}, \tag{5}$$

batches are ranked ascending by $\mathcal{C}(B_k, D_{\text{real}})$ and the top-ranked union of $K \cdot N_r$ samples constitutes D_{syn}. The representation map φ is inherited from Stage II when guidance is enabled, and defaults to the identity mapping when guidance is disabled.

Equation 5 and Equation 4 jointly minimize the same Chamfer functional $\mathcal{C}$ on the same representation space φ at two complementary operational levels: continuous gradient descent during reverse diffusion (Stage II) concentrates probability mass near the real-data manifold; discrete batch ranking after diffusion (Stage III) prunes off-manifold residuals. This joint minimization is the BCR principle; a saturation argument in K is given in Appendix A.

Alternative samplers. The grid search additionally exposes four alternative samplers (Stratified, IBOSS [38], HDBSCAN [7], MDSampler), which differ from the Chamfer Sampler in their selection criterion: feature-space stratification, information-theoretic boundary coverage, density-based clustering, and one-directional manifold-distance ranking. Their inclusion enables the grid to recover single-stage variants when Stage II disables guidance, in which case the Chamfer functional's bidirectional structure is no longer matched by Stage II and a one-directional alternative may suffice. Full definitions are deferred to Appendix D.

Generative Knowledge Distillation. GKD [17, 26] replaces the hard targets in D_{cand} with soft-label distributions produced by a teacher classifier f_{GKD} trained on D_{real}, yielding the relabeled candidate set $D_{\text{cand}}^{\text{GKD}}$; features are left unchanged (Appendix E). GKD is designed for classification targets but remains exposed to the grid search on regression benchmarks. The grid search treats GKD as an orthogonal toggle: each sampler is evaluated on both D_{cand} and $D_{\text{cand}}^{\text{GKD}}$, and the higher-utility variant is retained.

4 Experiments and Results

Datasets and configurations. We evaluate TARDIS on 15 publicly available tabular benchmarks (UCI and Kaggle) spanning binary classification (5 datasets), multiclass classification (4), and regression (6), with structural diversity across feature counts (8 to 46), feature types, instance counts (666 to 53,940), class balance, and domain coverage (healthcare, finance, environment, education, social science). All datasets use a 90/5/5 train/validation/test split seeded identically across models. The TPE search and inner grid of Section 3 are run independently per dataset; per-dataset structural attributes and the configurations selected by the search are reported in Appendix F (Tables 6 and 8, respectively), while Appendix M follows the configuration selected for Diabetic Retinopathy through the three stages.

Backbone selection. TabDiff [31] is selected as the backbone via a controlled 1:1 cardinality comparison against TabSyn [43] and TabularARGN [33] on the 10 benchmarks of our corpus shared with prior tabular diffusion work. Aggregating the per-dataset relative deviation from real-data utility on the headline metric, TabDiff is closest to Real with $\Delta_{\%} = -6.7\%$, followed by TabSyn (-8.9%) and TabularARGN (-25.7%). Per-dataset breakdowns and full metric coverage are deferred to Appendix B.

Evaluation protocol. The downstream learner is XGBoost [10]. The validation split serves both its early stopping and the validation utility maximized by the configuration search, and the test split serves only the reported evaluation; Appendix K repeats the evaluation on the six smallest corpora under a sealed protocol whose holdout is never seen by the search. We adopt the train-on-synthetic-test-on-real (TSTR) protocol, reporting AUROC for binary classification, weighted-AUROC for multiclass classification, and RMSE for regression as the primary downstream metric ψ. We report the per-dataset relative improvement $\Delta\%$ in ψ both over the real-data baseline (sign-corrected for RMSE) and over the TabDiff backbone, and aggregate it across the 15 datasets by median and mean, with percentile-bootstrap 95% confidence intervals on the mean; statistical significance is assessed by Wilcoxon signed-rank paired tests, with per-dataset means as the unit of comparison. Fidelity (Precision) and diversity (Recall) are reported via manifold metrics [23]; privacy proxies via Distance to Closest Record (DCR) and Nearest-Neighbor Distance Ratio (NNDR). Per-dataset and per-metric breakdowns are deferred to Appendix G. Unless stated otherwise, all experiments run on a single Nvidia GTX 1080 Ti (12 GB, 2017 consumer GPU), with 5 runs per configuration; the downstream learner is deterministic.

4.1 Main results

Aggregate downstream utility. Across the 15 benchmarks (Table 1), TARDIS achieves a median $\Delta\%$ improvement of $+8.6$ over $\mathcal{U}_{\text{real}}$ (mean $+9.8$, 95% CI of the mean $[+3.8, +16.4]$, Wilcoxon

Table 1: Per-dataset downstream utility under TSTR with XGBoost. Primary metric ψ: AUROC for BC, weighted-AUROC for MC (↑), RMSE for R (↓). TARDIS$_{K=1}$ is the full pipeline constrained to emit N_r rows selected from the same M=50 candidate pool, not a guidance-only variant (Appendix I). Δ%R and Δ%T are sign-corrected relative changes vs. Real and TabDiff. Best per row in bold; aggregates over the 15 datasets at the bottom.

Dataset	Real	TabDiff	TARDIS$_{K=1}$	TARDIS	Δ%R	Δ%T
Regression — RMSE ↓						
Solar Flare	0.1025	0.0923	0.0869	**0.0620**	+39.5	+32.8
Infrared Therm.	0.0541	0.0585	0.0562	**0.0482**	+10.9	+17.6
Abalone	0.081	0.079	0.075	**0.074**	+8.6	+6.3
Insurance	0.118	0.132	0.143	**0.110**	+6.8	+16.7
Beijing	**0.033**	0.051	0.051	0.037	-12.1	+27.5
News	0.01154	0.01194	0.01000	**0.00961**	+16.7	+19.5
Binary classification — AUROC ↑						
Diabetic Retin.	0.6976	0.6671	0.6697	**0.9037**	+29.5	+35.5
Adult	**0.927**	0.913	0.922	0.924	-0.3	+1.2
Default	0.764	0.765	0.768	**0.775**	+1.4	+1.3
Magic	0.956	0.940	0.949	**0.965**	+0.9	+2.7
Bank	0.929	0.856	0.863	**0.933**	+0.4	+9.0
Multiclass classification — weighted-AUROC ↑						
Student Perf.	0.7392	0.7453	0.6796	**0.8939**	+20.9	+19.9
Contraceptive	0.6491	0.6838	0.677	**0.7077**	+9.0	+3.5
Music	0.729	0.823	0.833	**0.840**	+15.2	+2.1
Diamonds	**0.943**	0.930	0.934	**0.943**	0.0	+1.4
Mean					+9.8	+13.1
Median					**+8.6**	+9.0
95% CI (mean)					[+3.8, +16.4]	[+7.5, +19.2]
p (Wilcoxon)					0.008	$< 10^{-4}$
Strict wins					12/15	**15/15**

$p = 0.008$), with strict positive improvement on 12 of 15 datasets and parity on Diamonds. Against the TabDiff backbone, the improvement is uniform: $\Delta\% > 0$ on every dataset, with median $+9.0$ and mean $+13.1$ (95% CI $[+7.5, +19.2]$, $p < 10^{-4}$). The proximity of mean and median over $\mathcal{U}_{\text{real}}$ indicates a tight distribution of dataset-level gains rather than an aggregate driven by outliers.

Cardinality ablation. Setting $K = 1$ constrains TARDIS to emit N_r rows, matching the cardinality of D_{real}, while leaving the $M = 50$ candidate pool and the Stage III selection in place. This variant (TARDIS$_{K=1}$, Table 1) improves over the TabDiff backbone on 11 of 15 datasets with one tie, at mean $\Delta\% = +1.2$, but does not significantly improve over $\mathcal{U}_{\text{real}}$ (Wilcoxon $p > 0.05$). Removing the selection step as well, by generating a single candidate per output row ($M = K = 1$), improves over the backbone on three of the nine datasets where guidance is enabled and falls behind TARDIS$_{K=1}$ on six of them (Appendix I). Closing the gap to real data therefore requires the cardinality dimension: candidate redundancy at Stage I and discrete refinement at Stage III carry an effect that Stage II guidance alone does not produce.

Utility inverse scaling. TARDIS's gain over $\mathcal{U}_{\text{real}}$ shrinks as the real-data baseline approaches the metric ceiling (Figure 2): the four datasets where the real baseline approaches the metric ceiling (Adult AUROC 0.927, Bank 0.929, Magic 0.956, Diamonds 0.943) yield $\Delta\% \in [-0.3, +0.9]$, with Adult's -0.3 the only negative point. Beijing represents the analogous case for regression: the target admits a near-linear structure that XGBoost on real data already exploits, leaving a residual deviation of -12.1. Improvement concentrates instead on datasets where $\mathcal{U}_{\text{real}}$ reflects data scarcity, label noise, or class imbalance: Solar Flare ($+39.5$, 1,066 instances), Diabetic Retinopathy ($+29.5$, 1,151 instances), Student Performance ($+20.9$, 666 instances), News ($+16.7$, imbalanced regression target), Music ($+15.2$, multiclass with rare classes).

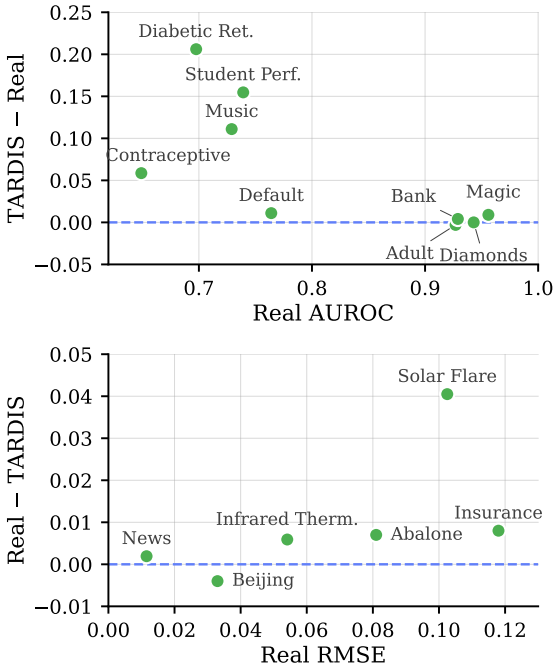


Figure 2: Per-dataset difference between TARDIS and $\mathcal{U}_{\text{real}}$ (sign-corrected for RMSE) as a function of the real-data baseline, classification (top, AUROC) and regression (bottom, RMSE). Gains concentrate on datasets with a weak real-data baseline.

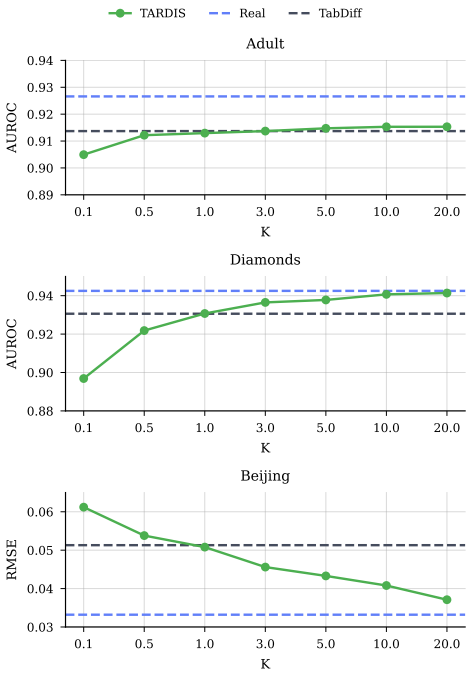


Figure 3: Cardinality saturation: sweep of K with $M = 50$ on Adult, Diamonds, and Beijing, the three benchmarks where TARDIS yields the smallest gain over Real (Table 1). Curves report the configuration with GKD disabled. TARDIS still overtakes TabDiff near $K \approx 1.0$ on all three.

4.2 Empirical signatures of BCR

Cardinality saturation. We sweep $K \in \{0.1, 0.5, 1, 3, 5, 10, 20\}$ with $M = 50$ fixed on three benchmarks spanning the $\mathcal{U}_{\text{real}}$ regime: Adult (AUROC 0.927), Diamonds (AUROC 0.943), and Beijing (RMSE 0.033). With GKD disabled, all three datasets exhibit monotone improvement in ψ (Figure 3), with knees at $K \approx 0.5$ on Adult and $K \approx 5$ on Diamonds; on Beijing the curve continues to descend through $K = 20$ without plateauing within the swept range. TARDIS overtakes TabDiff near $K \approx 1.0$ on all three datasets, indicating that BCR refinement begins to add value already at real cardinality. The plateau location is ordered consistently with the strength of $\mathcal{U}_{\text{real}}$: Adult saturates early below Real, Diamonds reaches parity with Real near $K = 20$, and Beijing remains below Real within the swept range.

Sampler-guidance coupling. The TPE search and inner grid expose a structural alignment between Stage II guidance and Stage III sampler choice, recovered without explicit constraint (Table 8). Of the 9 datasets where guidance is enabled, all 9 select the Chamfer Sampler in Stage III; of the 6 datasets where guidance is disabled, 4 (Bank, Magic, Music, Abalone) select the MDSampler, and the remaining 2 (Adult, Solar Flare) select the Chamfer Sampler with GKD enabled (the only configurations with $K \leq 1$, where too few rows are retained for the MDSampler's purely fidelity-driven selection to preserve coverage). The configurations therefore cluster into two coherent regimes: guided pool with bidirectional Chamfer-based selection $(9/15)$, and unguided pool with one-directional manifold-distance selection $(4/15)$. When Stage II shapes the candidate pool toward the real manifold via $\nabla\mathcal{C}$, the discrete sampler that minimizes the same $\mathcal{C}$ dominates; when Stage II is disabled and $K \gg 1$, the simpler one-directional MDSampler suffices.

Low-cardinality regime. The configuration-level evidence in Table 8 surfaces a regime separation along K that we term the *low-cardinality regime* ($K \leq 1$). Across the 15 benchmarks, GKD is enabled on 2 datasets only, both with $K \leq 1$ (Adult K=0.5, Solar Flare K=0.1); on the remaining 13,

all with $K \geq 8.5$, the search disables GKD. The pattern is consistent with the qualitative mechanism: at low K the Chamfer Sampler retains too few rows to recover utility through selection alone, and the soft-label channel becomes the dominant signal; at high K selection suffices, and GKD's smoothing becomes a net negative. Although designed for classification, GKD remains selectable on regression targets, and its only regression selection (Solar Flare, K=0.1) falls within the same regime. We position GKD as a regime-specific component for hardware-constrained downstream applications.

4.3 Fidelity, diversity, privacy, and runtime

Fidelity and diversity. Manifold Precision (fidelity) and Recall (diversity) [23] are preserved at the TabDiff backbone level on all 15 datasets (Table 2): mean Precision 0.949 (TARDIS) vs. 0.947 (TabDiff), mean Recall 0.995 vs. 0.994, with per-dataset absolute deviations bounded by 0.043 and 0.003 respectively. The BCR-driven refinement does not collapse the synthetic distribution onto a subset of the real manifold, the canonical failure mode of utility-aware curation.

Table 2: Fidelity, diversity, and privacy proxies across 15 datasets.

Metric	TabDiff	TARDIS
Precision ↑	0.947 ± 0.073	$\mathbf{0.949 \pm 0.069}$
Recall ↑	0.994 ± 0.005	$\mathbf{0.995 \pm 0.005}$
DCR-1 share	$\mathbf{0.501 \pm 0.010}$	$\mathbf{0.499 \pm 0.013}$
NNDR ↑	0.823 ± 0.115	$\mathbf{0.824 \pm 0.107}$

Privacy. DCR-1 share and NNDR, which measure proximity to training records, are likewise preserved at the backbone level (Table 2; per-dataset breakdown in Table 10): TARDIS DCR-1 share 0.499 ± 0.013 vs. TabDiff 0.501 ± 0.010, NNDR 0.824 ± 0.107 vs. 0.823 ± 0.115, with no per-dataset deviation exceeding 0.041. A loss-based membership-inference attack [42] on XGBoost reaches a mean AUC of 0.542 when the learner is trained on the TARDIS synthetic dataset, 0.520 on the TabDiff one, and 0.765 on the real records (Appendix G.3). TARDIS does not improve privacy (which is not a claim of this paper), and the selection costs about 0.02 attack AUC relative to the backbone: although the Chamfer Sampler ranks candidates by proximity to D_{real}, the bidirectional structure of $\mathcal{C}$ balances fidelity against coverage, and exact matches with training records remain at the backbone level.

Runtime. On the Nvidia GTX 1080 Ti, TARDIS adds 0.9 to 79.4 minutes of wall-clock time on top of the trained backbone to apply the selected configuration, excluding the one-time configuration search, with a median of 16.1 minutes across the 15 datasets; the largest case (Bank, $\sim 45{,}000$ instances) runs in 79.4 minutes and the smallest (Solar Flare) in 0.9 minutes (per-stage breakdown in Appendix H). Stage II generation accounts for 47% to 90% of the wall-clock budget; Stage III selection contributes 5% to 38%, with the highest fractions on Music and News (both 38%); GKD distillation contributes 7% to 11% on the two datasets where it is active. The dominant cost factor is the $M = 50$ oversampling. On an NVIDIA L40S the median cost falls to 3.6 minutes, 1.8% of the time needed to train the TabDiff backbone on the same GPU and at most 6.8% (Appendix H).

5 Discussion and Limitations

BCR as a domain-transferable principle. The Chamfer functional of Equation 2 is modality-agnostic: it requires only a differentiable representation map φ and a real-data reference batch, so both of its operationalizations can be instantiated for image, text, and joint multimodal diffusion under domain-appropriate encoders, since neither a differentiable score nor a candidate pool is tabular-specific. What our experiments establish is a two-part claim. TARDIS is an inference-time search framework over the space of Section 3, and BCR is the region of that space the unconstrained per-dataset search elects in 9 of 15 cases (Section 4.2). Within that region the discrete level carries the dominant share of the gap-closing effect: at matched output size the pipeline improves over the backbone but not over real data, and removing the selection step as well leaves gains on a minority of datasets (Section 4.1, Appendix I). This inverts the image-domain emphasis on score-level guidance. Proximity to training records stays at the backbone level (Section 4.3), consistent with the coverage term penalizing collapse onto a subset of D_{real}; since $\mathcal{C}$ vanishes when the synthetic set coincides with D_{real}, this is an empirical finding, not a guarantee. Refinement is therefore complementary to backbone improvement, not a substitute, with gains concentrated where synthetic data is most needed (small corpora, label noise, class imbalance).

Transfer and attribution. Stage III can be applied to any candidate pool: on TabSyn pools, without retraining, the Chamfer Sampler at the selected cardinality $K \cdot N_r$ improves over the standard TabSyn

sample of N_r rows on eight corpora (median $\Delta\% = +1.7$; Appendix B). The synthetic datasets, searched with XGBoost, also transfer zero-shot to other learners: an MLP improves over its real-data counterpart on ten corpora (median $\Delta\% = +2.1$), while CatBoost and linear models remain at parity, consistent with linear models being insensitive to the local manifold structure that the refinement restores (Appendix L). Changing one component at a time attributes the benefit to cardinality and selection: at fixed selection and cardinality, guidance is not distinguishable from pool resampling, and the pool multiplier M yields no consistent benefit (Appendix J).

Choosing M and K. The benefit is carried by K, whereas M yields no consistent benefit and sets most of the cost (Appendix J.1, Section 4.3). We therefore recommend setting K within the cardinality budget of the downstream application and M just above K. Under a budget of $K \approx 1$, GKD can be enabled on classification; near the metric ceiling, parity with real data is the expected outcome and the configuration without guidance is cheaper (Appendix J.4). A fixed Stage III configuration (Chamfer selection at $K = 10$, no GKD) on the same candidate pools is a fallback, which the full search outperforms on 14 of 15 datasets (Appendix J.3).

Limitations and Future Work. The full pipeline is validated on a single backbone (TabDiff). Stage II requires a continuous reverse process: it is untested on TabSyn [43] and TabDDPM [22] and does not apply to autoregressive (TabularARGN [33]) or variational (TVAE [40]) decoders. Applied alone, Stage III yields no consistent gain on CTAB-GAN+ [44] and TVAE pools, consistent with selection requiring a candidate pool that already covers the real-data manifold (Appendix B). The configurations are searched with XGBoost as validation utility, and other learners are evaluated zero-shot on ten of the fifteen corpora (Appendix L). The pool multiplier is fixed at $M = 50$ and swept only at $K \in \{1, 10\}$ on eight corpora (Appendix J.1), so the joint dependence on M and K is not characterized. The downstream learner and the samplers of the headline configurations are deterministic, so run-to-run variance reflects pool generation only. Privacy is assessed through proximity proxies and a loss-based membership-inference attack (Appendix G.3), without formal differential-privacy guarantees. Future work targets transfer to TabDDPM and to image-domain diffusion, a joint search over M and K that accounts for N_r, the manifold-coverage gap, and the deployment cardinality budget, and a differentially private variant that spends a calibrated privacy budget at both refinement levels.

6 Conclusions

We introduced TARDIS, an inference-time search framework for pre-trained tabular diffusion backbones, organized around the *Bidirectional Chamfer Refinement* (BCR) principle: a single Chamfer functional minimized continuously during reverse diffusion via score-level guidance and discretely after sampling via batch ranking. BCR is the region of the search space that the per-dataset search elects on 9 of 15 benchmarks, with single-stage variants on the remaining 6. Across 15 heterogeneous tabular benchmarks (binary, multiclass, regression) and at its selected cardinality, TARDIS achieves a median $+8.6\%$ improvement in ψ over models trained on real data (Wilcoxon $p = 0.008$, $12/15$ strict wins) and improves over the TabDiff backbone on every dataset (mean $+13.1\%$, $p < 10^{-4}$), with fidelity, diversity, and privacy proxies preserved at the backbone level and an apply-time overhead under 80 minutes on a 2017 consumer GPU. The synthetic–real gap is therefore not primarily a training-time problem on the studied corpus, and BCR provides a domain-transferable principle for inference-time refinement of pre-trained diffusion backbones.

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A Bidirectional Chamfer Refinement: Properties and Saturation

A.1 Properties of the Symmetric Chamfer Functional

Let $\varphi : \mathcal{X} \to \mathbb{R}^d$ be a representation map and $A, B \subset \mathcal{X}$ finite multisets with φ injective on $A \cup B$. The functional $\mathcal{C}$ of Equation 2 satisfies: **(P1) non-negativity**, $\mathcal{C}(A, B) \geq 0$ (Euclidean norm under uniform weighting); **(P2) identity of indiscernibles**, $\mathcal{C}(A, B) = 0 \iff \varphi(A) = \varphi(B)$ as multisets, since fidelity and coverage terms vanish jointly only when the two sets coincide in φ-space; **(P3) symmetry**, $\mathcal{C}(A, B) = \mathcal{C}(B, A)$ by construction. P1–P3 jointly justify the use of $\mathcal{C}$ as a single optimization target: minimizing $\mathcal{C}(D_{\text{syn}}, D_{\text{real}})$ at the continuous (Stage II) and discrete (Stage III) refinement levels is a coherent objective, since any reduction at one level corresponds to monotone movement toward the same target distribution in φ-space.

A.2 The BCR Saturation Argument

We argue that, under reasonable assumptions on the candidate pool and the discrete sampler, $\mathcal{U}_{\text{syn}}$ exhibits a monotone-then-plateau dependence on the cardinality multiplier K, with a dataset-dependent knee location. We state this as a principled argument rather than a formal theorem: we do not characterize the convergence rate as $M \to \infty$, the empirical knee depends on the intrinsic dimension of D_{real} in φ-space which we do not formalize, and all reported configurations hold $M = 50$ fixed.

Argument (BCR Saturation). Under the assumptions that (i) Stage II concentrates the empirical density of $\varphi(D_{\text{cand}})$ around the support of $\varphi(D_{\text{real}})$, (ii) Stage III selects the top K/M fraction of D_{cand} by $\mathcal{C}$-rank, and (iii) the downstream learner is consistent in the cardinality of training data, we expect $\mathcal{U}_{\text{syn}}$ to be monotonically non-decreasing in K and to saturate at $K = K^*$ for a dataset-dependent knee $K^* \leq M$.

Reasoning. As K decreases relative to M, the Stage III sampler retains only the most $\mathcal{C}$-aligned candidates, sharpening $\varphi(D_{\text{syn}})$ around the support of $\varphi(D_{\text{real}})$ and reducing the fidelity term m_{sr} of Equation 5; as $K \to M$, the selection becomes vacuous and the discrete refinement contributes negligibly. Saturation occurs when the coverage term m_{rs} becomes the binding constraint, i.e., when a smaller selection fraction risks under-covering isolated regions of the real-data manifold. The knee K^* is expected to scale with the effective intrinsic dimension of D_{real} in φ-space and with the inverse local density of real samples.

A.3 Empirical Validation

The cardinality sweep $K \in \{0.1, 0.5, 1, 3, 5, 10, 20\}$ with $M = 50$ on Adult, Diamonds, and Beijing (Figure 3) is consistent with the predicted monotone-then-plateau profile. The knees occur at $K^* \approx 0.5$ for Adult, $K^* \approx 5$ for Diamonds, and beyond $K = 20$ for Beijing, an ordering consistent with the intrinsic-dimension argument of Section A.2. The sweeps of the pool multiplier in Appendix J.1 support reading the knee in terms of K alone: M yields no consistent benefit at $K = 1$, and at $K = 10$ its spread is of the order of the seed-to-seed standard deviation.

B Backbone Selection

We select TabDiff [31] as backbone from a comparison against TabSyn [43] and TabularARGN [33] on the 10 corpora of Table 3, where all three generators are trained at matched 1:1 cardinality and evaluated under the TSTR protocol of Section 4 with several utility and fidelity metrics per dataset.

Aggregate dominance. Of the 50 metric cells, TabDiff dominates on 23, TabSyn on 14, and TabularARGN on 13, with dominance concentrated on the primary TSTR metrics (classification AUROC, regression RMSE).

Architecture and TARDIS compatibility. TabDiff's per-feature noise schedules differentiate noising rates across features of heterogeneous scale and cardinality, avoiding the bias that a single shared schedule introduces on mixed-type inputs; its continuous reverse process and exposed score ε_θ make it directly compatible with the Stage II score perturbation of Equation 4. TabSyn is similarly compatible via latent-space score perturbation, but loses the per-feature inductive bias.

Table 3: Backbone comparison on 10 corpus benchmarks under matched 1:1 cardinality. Primary downstream metric reported: AUROC ↑ for binary classification, weighted-AUROC ↑ for multiclass classification, RMSE ↓ for regression. Bold indicates the best generator per column, excluding Real. The final column reports the average percentage delta with respect to Real, computed as $100(m-r)/r$ for AUROC and $100(r-m)/r$ for RMSE; higher is better.

	Classification (AUROC ↑)						Regression (RMSE ↓)				$\Delta_{\%}$ vs Real ↑
Model	Adult	Default	Magic	Bank	Music	Diamonds	Abalone	Insurance	Beijing	News	Avg.
Real	.927	.764	.956	.929	.729	.943	.081	.118	.033	.01154	–
TabularARGN	**.914**	.751	.909	**.911**	.714	.907	.096	.281	.062	**.01117**	−25.72
TabSyn	.908	**.768**	**.942**	.896	.724	.929	**.079**	**.132**	.055	.01201	−8.88
TabDiff	.913	.765	.940	.856	**.823**	**.930**	**.079**	**.132**	**.051**	.01194	**−6.68**

Transfer of the discrete level. Stage II requires a continuous reverse process, whereas Stage III applies to any candidate pool. We therefore apply the Chamfer Sampler alone to the archived candidate pools of three further generators: TabSyn at $M = 50$, and CTAB-GAN+ and TVAE at $M = 10$. No generator is retrained and no guidance is applied; the sampler retains $K \cdot N_r$ rows with the K of Table 8, capped at $K = 9$ on the $M = 10$ pools, and XGBoost is evaluated under the TSTR protocol of Section 4 against the standard sample of the same generator (N_r rows). Values are means over 4 seeds on NVIDIA L40S GPUs.

Latent-space diffusion. On TabSyn (Table 4), Chamfer selection of $K \cdot N_r$ rows improves over the standard sample of N_r rows on all eight corpora of the table (median $\Delta\% = +1.7$), with the largest gains where the standard sample is weakest (Beijing $+19.5$, News $+10.7$, Insurance $+8.9$).

GAN and VAE pools. On CTAB-GAN+ and TVAE (Table 5) the gains vanish (median $\Delta\% = +0.1$, range $[-3.0, +4.8]$), consistent with selection requiring a candidate pool that already covers the real-data manifold.

Table 4: Stage III Chamfer selection on TabSyn candidate pools ($M = 50$), without retraining and without guidance. AUROC and weighted-AUROC ↑, RMSE ↓. $\Delta\%$ is the sign-corrected relative change with respect to the standard TabSyn sample (N_r rows).

Dataset	Metric	TabSyn	TabSyn + Stage III	$\Delta\%$
Magic	AUROC	0.9422	0.9560	+1.5
Insurance	RMSE	0.1321	0.1203	+8.9
Adult	AUROC	0.9080	0.9084	+0.0
Default	AUROC	0.7675	0.7705	+0.4
Beijing	RMSE	0.0553	0.0446	+19.5
Diamonds	AUROC	0.9287	0.9393	+1.1
Bank	AUROC	0.8956	0.9122	+1.9
News	RMSE	0.0120	0.0107	+10.7
Median				+1.7

Table 5: Stage III Chamfer selection on CTAB-GAN+ and TVAE candidate pools ($M = 10$), without retraining. AUROC ↑, RMSE ↓. $\Delta\%$ is the sign-corrected relative change with respect to the standard sample of the same generator (N_r rows).

Dataset	Metric	Generator	Generator + Stage III	$\Delta\%$
CTAB-GAN+				
Adult	AUROC	0.9031	0.9019	−0.1
Default	AUROC	0.7451	0.7517	+0.9
Beijing	RMSE	0.0732	0.0697	+4.8
News	RMSE	0.01093	0.01089	+0.4
TVAE				
Adult	AUROC	0.8881	0.8833	−0.5
Beijing	RMSE	0.0782	0.0806	−3.0
Median				+0.1

C Stage II Schedules and Reference Batching

All schedule families and batching strategies below are exposed to the TPE search; the bounds $\gamma_{\min}/\gamma_{\max}$ are searched per dataset over the domains of Table 7, and $\beta_{\min}/\beta_{\max}$ are selected per dataset.

C.1 Guidance Schedules $\Gamma(t)$

Let T denote the total number of reverse-diffusion steps, with $t \in [0, T]$ counting from $t = T$ (pure noise) to $t = 0$ (final sample). We define four functional families for $\Gamma(t)$, each parameterized by $(\gamma_{\min}, \gamma_{\max})$ and bounded in $[\gamma_{\min}, \gamma_{\max}]$:

$$\textbf{Constant:} \quad \Gamma(t) = \gamma_{\max} \tag{6}$$

$$\textbf{Linear:} \quad \Gamma(t) = \gamma_{\min} + (\gamma_{\max} - \gamma_{\min}) \cdot \frac{t}{T} \tag{7}$$

$$\textbf{Cosine:} \quad \Gamma(t) = \gamma_{\min} + \frac{1}{2} \cdot (\gamma_{\max} - \gamma_{\min}) \cdot \left(1 + \cos\left(\pi \cdot \frac{t}{T}\right)\right) \tag{8}$$

$$\textbf{Sine:} \quad \Gamma(t) = \gamma_{\max} - \frac{1}{2} \cdot (\gamma_{\max} - \gamma_{\min}) \cdot \left(1 + \cos\left(\pi \cdot \frac{t}{T}\right)\right) \tag{9}$$

C.2 KL Annealing Schedules $\beta(t)$

Let T_{VAE} denote the total number of training epochs of the β-VAE encoder φ. We define five functional families for the KL weight $\beta(t)$, each parameterized by $(\beta_{\min}, \beta_{\max})$:

$$\textbf{Constant:} \quad \beta(t) = \beta_{\max} \tag{10}$$

$$\textbf{Linear:} \quad \beta(t) = \beta_{\max} \cdot \min\left(1, \frac{t}{T_{\text{VAE}}}\right) \tag{11}$$

$$\textbf{Cosine:} \quad \beta(t) = \beta_{\min} + \frac{1}{2} \cdot (\beta_{\max} - \beta_{\min}) \cdot \left(1 + \cos\left(\pi \cdot \frac{t}{T_{\text{VAE}}}\right)\right) \tag{12}$$

$$\textbf{Sine:} \quad \beta(t) = \beta_{\max} - \frac{1}{2} \cdot (\beta_{\max} - \beta_{\min}) \cdot \left(1 + \cos\left(\pi \cdot \frac{t}{T_{\text{VAE}}}\right)\right) \tag{13}$$

$$\textbf{Cyclical:} \quad \beta(t) = \beta_{\max} \cdot \frac{t \bmod T_{\text{cyc}}}{T_{\text{cyc}}} \tag{14}$$

The Cyclical schedule [15] alternates between regularized and reconstruction-emphasized regimes to mitigate posterior collapse on high-cardinality categorical features.

C.3 Reference Batching

The reference batch x_r used in Equation 4 is sampled at the start of each generation and held fixed across all reverse-diffusion steps.

Class-conditional batching. For classification, x_r is drawn from the target class of the candidate batch x_t, concentrating $\nabla_{x_t}\mathcal{C}$ on the class-conditional manifold.

Global batching. x_r is drawn uniformly from D_{real}; this is the default for regression and an option for classification.

D Stage III Sampler Definitions

The five samplers of the Stage III grid take the candidate pool D_{cand} of cardinality $M \cdot N_r$ produced by Stage II and return the synthetic dataset D_{syn} of cardinality $K \cdot N_r$.

D.1 Chamfer Sampler

The Chamfer Sampler (Section 3.3) partitions D_{cand} into batches $\{B_k\}$ of size b, ranks each by $\mathcal{C}(B_k, D_{\text{real}})$ (Equation 5), and retains the union of the top-ranked batches up to $K \cdot N_r$ samples.

D.2 Stratified Sampler

The Stratified Sampler partitions D_{cand} along the target variable for classification tasks, or along quantile bins of the target for regression, and samples uniformly within each stratum to match the per-class frequencies of D_{real}. The selection is independent of feature-space proximity, retaining only the marginal target distribution of the real set.

D.3 IBOSS Sampler

IBOSS (Information-Based Optimal Subdata Selection) [38] selects candidates that span the feature space, ensuring every region of the candidate manifold has comparable representation in D_{syn}. It favors extremal points at the boundary of the candidate distribution and is target-agnostic, operating independently of D_{real}.

D.4 HDBSCAN Sampler

The HDBSCAN Sampler clusters D_{cand} in φ-space using the HDBSCAN algorithm [7] with $\text{min_cluster_size} = b$, retaining the top $K \cdot N_r$ clusters by density. Clusters are selected without reference to D_{real}.

D.5 MDSampler

The MDSampler (Manifold-Distance Sampler) approximates the support of $\varphi(D_{\text{real}})$ as the union of k-nearest-neighbor *hyperspheres* around each real point in φ-space, with each *hypersphere* centered at $\varphi(r)$ and radius $\rho_k(r)$ equal to the distance from $\varphi(r)$ to its k-th nearest neighbor in $\varphi(D_{\text{real}})$ (we use $k = 10$). Each candidate $x \in D_{\text{cand}}$ is scored by its distance to this estimated manifold,

$$d_{\text{MD}}(x) = \min_{r \in D_{\text{real}}} \max\big(0,\, \|\varphi(x) - \varphi(r)\|_2 - \rho_k(r)\big)\,, \tag{15}$$

which is zero whenever x lies inside any *hypersphere* and equals the distance from the nearest *hypersphere* boundary otherwise. The MDSampler retains the $K \cdot N_r$ candidates with smallest d_{MD}, prioritizing candidates inside the real-data manifold and falling back on the closest off-manifold ones once the inside set is exhausted. The construction follows the manifold Precision proxy of Kynkäänniemi et al. [23]: $d_{\text{MD}} = 0$ corresponds to the inside-the-manifold criterion that defines that metric. The MDSampler is the fidelity-driven counterpart of the bidirectional Chamfer Sampler: when Stage II is disabled, the candidate pool retains broad coverage by construction, the coverage term m_{rs} of Equation 5 is approximately satisfied a priori, and inside-manifold selection suffices; when Stage II is enabled and concentrates the pool around $\varphi(D_{\text{real}})$, most candidates already satisfy $d_{\text{MD}} \approx 0$ and the bidirectional Chamfer Sampler is needed to discriminate via coverage.

E Generative Knowledge Distillation Details

Procedure. GKD trains a teacher classifier $f_{\text{GKD}} : \mathcal{X} \to \Delta^{|\mathcal{Y}|-1}$ on D_{real}, where $\Delta^{|\mathcal{Y}|-1}$ denotes the $(|\mathcal{Y}| - 1)$-dimensional probability simplex. For each candidate $\hat{x}_j \in D_{\text{cand}}$ produced by Stage II, the hard target $\hat{y}_j$ is replaced by the soft-label distribution $f_{\text{GKD}}(\hat{x}_j)$, yielding the relabeled set

$$D_{\text{cand}}^{\text{GKD}} = \big\{(\hat{x}_j,\, f_{\text{GKD}}(\hat{x}_j))\big\}_{j=1}^{|D_{\text{cand}}|}\,, \tag{16}$$

which is then passed to Stage III's sampler in place of D_{cand}. The downstream learner A trains under cross-entropy with soft targets. GKD relabels targets only, leaving feature vectors $\hat{x}_j$ unchanged.

Teacher classifier. In our experiments f_{GKD} is XGBoost trained on D_{real} under the same hyperparameter search as the downstream learner A, with predicted-probability outputs used as soft labels.

Regression targets. Soft labels over $\mathbb{R}$ would require a parametric assumption on the conditional distribution, so on regression targets GKD replaces each target with the prediction of the teacher f_{GKD}. Among the six regression datasets, the search selects it only on Solar Flare ($K = 0.1$).

F Per-Dataset Configurations

F.1 Dataset Summary

Table 6 reports the structural attributes of the 15 corpus benchmarks, sourced from UCI and Kaggle. Three datasets contain missing values (Adult, Beijing, Infrared Thermography), imputed identically across all models: median for numerical features, mode for categorical, per training split.

Table 6: Dataset summary. n = number of instances; d = number of feature columns excluding the target; d_n/d_c = numerical / categorical feature counts; NaN indicates presence of missing values (imputed identically across all models).

Dataset	Task	n	d	d_n	d_c	NaN
Binary classification						
Adult [3]	BC	48,842	14	6	8	Yes
Bank [27]	BC	45,211	16	7	9	No
Default [41]	BC	30,000	23	14	9	No
Diabetic Retinopathy [2]	BC	1,151	19	19	0	No
Magic [5]	BC	19,020	10	10	0	No
Multiclass classification						
Contraceptive [25]	MC (3)	1,473	9	2	7	No
Diamonds [32]	MC (5)	53,940	9	6	3	No
Music [30]	MC (11)	25,709	14	13	1	No
Student Performance [36]	MC (4)	666	11	0	11	No
Regression						
Abalone [28]	R	4,177	8	7	1	No
Beijing [9]	R	43,824	11	6	5	Yes
Infrared Thermography [39]	R	1,020	33	30	3	Yes
Insurance [37]	R	987	10	3	7	No
News [14]	R	39,644	46	44	2	No
Solar Flare [34]	R	1,066	10	7	3	No

Table 7: Stage II hyperparameter search space optimized per dataset by the Tree-structured Parzen Estimator. Domains follow Optuna [1] conventions: *Cat* = categorical, *Int* = integer-valued with the indicated step, *Log* = log-uniform, *Lin* = linear with the indicated step. The β-VAE inner study (latent dimension and KL annealing schedule) is invoked only when the outer search selects β-VAE mapping; its search space is reported in Appendix C.

Parameter	Domain	Description	Scale
K	$[0.1, 25.0]$	Final cardinality multiplier	Lin (0.1)
Guidance variant	{None, identity, β-VAE}	Toggle and form of φ	Cat
Schedule $\Gamma(t)$	{constant, linear, cosine, sine}	Guidance scaling family	Cat
$\gamma_{\max}$	$[10^{-1}, 5.0]$	Upper bound of $\Gamma(t)$	Log
$\gamma_{\min}$	$[10^{-8}, \gamma_{\max} - 0.1]$	Lower bound of $\Gamma(t)$	Log
t_g	$[3, 20]$	Guidance step count	Int (1)
Batching ratio	{None, $1, 1/2, 1/4, 1/8, 1/10$}	Class-conditional fraction	Cat

F.2 TPE Search Space

Table 7 lists the Stage II search space.

Table 8: Per-dataset configurations selected by the search procedure against the validation split. Stage II hyperparameters from TPE optimization; Stage III combination from grid search over $\{\text{GKD off}, \text{GKD on}\} \times \{\text{Stratified}, \text{IBOSS}, \text{HDBSCAN}, \text{MDSampler}, \text{Chamfer Sampler}\}$. Columns: K cardinality multiplier; G guidance toggle (✗ disabled, ✓ enabled); Γ schedule family (Section C.1); φ representation map (β-VAE(d) for β-VAE mapping with latent dimension d, identity for identity mapping, dash when guidance is disabled); B_{ratio} reference batching fraction; Type class-conditional (*cls*) or global (*glob*). CS = Chamfer Sampler, MD = MDSampler.

	Stage II (TPE)						Stage III (grid)	
Dataset	K	G	Γ	φ	B_{ratio}	Type	GKD	Sampler
Adult	0.5	✗	–	–	–	cls	✓	CS
Bank	24.9	✗	–	–	–	cls	✗	MD
Default	13.1	✓	cosine	identity	1/4	cls	✗	CS
Diabetic Retin.	22.5	✓	cosine	β-VAE(4)	1/2	cls	✗	CS
Magic	23.9	✗	–	–	–	cls	✗	MD
Contraceptive	8.5	✓	sine	β-VAE(16)	1/4	cls	✗	CS
Diamonds	23.1	✓	sine	β-VAE(4)	1/4	cls	✗	CS
Music	20.0	✗	–	–	–	cls	✗	MD
Student Perf.	15.2	✓	cosine	β-VAE(16)	1/8	cls	✗	CS
Abalone	23.0	✗	–	–	–	glob	✗	MD
Beijing	20.0	✓	cosine	identity	1	glob	✗	CS
Infrared Therm.	12.8	✓	cosine	β-VAE(32)	1/2	glob	✗	CS
Insurance	10.0	✓	sine	identity	1/10	glob	✗	CS
News	14.8	✓	constant	β-VAE(4)	1/8	glob	✗	CS
Solar Flare	0.1	✗	–	–	–	glob	✓	CS

F.3 Selected Per-Dataset Configurations

Aggregate selection counts. Of the 9 datasets enabling guidance, 3 select identity mapping and 6 β-VAE mapping, and all 9 select the Chamfer Sampler. Of the 6 datasets without guidance, 4 select the MDSampler and 2 the Chamfer Sampler with GKD, both at $K \leq 1$ (Adult, Solar Flare). Reference batching is class-conditional on all classification datasets and global on all regression datasets.

G Auxiliary Metrics

G.1 Classification F1

Table 9: Per-dataset auxiliary downstream classification metrics under TSTR: F1 for binary classification, macro-F1 for multiclass classification. Best per row in bold.

Dataset	Metric	Real	TabDiff	TARDIS$_{K=1}$	TARDIS
Diabetic Retin.	F1	0.5329	0.5469	0.5999	**0.7407**
Adult	F1	**0.707**	0.671	0.696	0.704
Default	F1	0.447	0.443	0.460	**0.460**
Magic	F1	0.862	0.817	0.837	**0.880**
Bank	F1	0.538	0.463	0.503	**0.562**
Student Perf.	macro-F1	0.5076	0.5119	0.3972	**0.7299**
Contraceptive	macro-F1	0.4229	0.4622	0.4671	**0.5322**
Music	macro-F1	0.364	0.470	0.471	**0.489**
Diamonds	macro-F1	0.801	0.774	0.787	**0.804**

F1. TARDIS has the best F1 on 8 of 9 classification datasets; the exception is Adult (0.707 Real vs. 0.704 TARDIS).

Table 10: Per-dataset DCR-1 share (closer to 0.50 indicates better balance between train-data proximity and novelty) and NNDR (↑).

	DCR-1 share		NNDR	
Dataset	TabDiff	TARDIS	TabDiff	TARDIS
Solar Flare	0.4906	0.5063	0.8876	0.8673
Infrared Therm.	0.5283	0.4902	0.9517	0.9540
Abalone	0.490	0.494	0.853	0.852
Insurance	0.487	0.487	0.762	0.773
Beijing	0.501	0.500	0.857	0.857
News	0.503	0.505	0.942	0.943
Diabetic Retin.	0.5068	0.5382	0.8706	0.8689
Adult	0.502	0.498	0.769	0.768
Default	0.505	0.500	0.822	0.823
Magic	0.501	0.491	0.882	0.879
Bank	0.489	0.497	0.911	0.910
Student Perf.	0.5008	0.4758	0.4970	0.5373
Contraceptive	0.5061	0.5023	0.6995	0.6871
Music	0.506	0.504	0.883	0.877
Diamonds	0.500	0.499	0.761	0.761

G.2 Per-Dataset Privacy Breakdown

Proximity to training records. TARDIS stays within ± 0.04 of TabDiff in DCR-1 share on all 15 datasets (within ± 0.02 on 12) and within ± 0.05 in NNDR; the largest deviations occur on small corpora.

G.3 Membership Inference and Near-Duplicates

Setting. The loss-based attack of Yeom et al. [42] targets the downstream XGBoost learner of Section 4: it scores each record by the loss of the learner, with the training records of the backbone, i.e., D_{real}, as members and the held-out records as non-members, each subsampled to at most $2{,}000$ records. We attack three instances of the learner, trained on D_{real} (Real), on the TARDIS synthetic dataset, and on the TabDiff synthetic dataset; the first corresponds to releasing a model trained on the real records instead of synthetic data. Each entry is a single run on NVIDIA L40S GPUs.

Membership inference. The learner trained on the real records exposes membership, markedly on small corpora (attack AUC up to 0.997, TPR@1%FPR up to 0.967; Table 11). Training on either synthetic dataset brings the attack close to chance level: mean attack AUC 0.542 for TARDIS and 0.520 for TabDiff vs. 0.765 for Real, and mean TPR@1%FPR 0.011 vs. 0.119.

Near-duplicates. A synthetic sample counts as an exact match when all its features coincide with a training record after rounding numerical features to six decimals; nearest-neighbor distances are Euclidean in z-scored one-hot space (Table 12). Exact matches are essentially zero wherever features are continuous; non-negligible fractions occur only on small, heavily discrete corpora (Solar Flare, Student Performance, Contraceptive), where collisions follow from the finite feature space. TARDIS remains at the backbone level (mean exact-match fraction 0.077 vs. 0.079, mean nearest-neighbor distance 1.50 vs. 1.42).

H Runtime Breakdown

Table 13 reports the per-stage wall-clock time of the refinement on the GTX 1080 Ti; it excludes TabDiff training, performed once per dataset and reused across configurations.

Current hardware and training cost. Table 14 re-measures the apply-time cost (Stage II, Stage III, and GKD where enabled) on an NVIDIA L40S from the logs of full pipeline reruns, which select a different Stage III sampler than Table 8 on 9 of the 15 datasets, and adds the training time of TabDiff (8000 epochs) and the end-to-end time of a TARDIS run on the same GPU.

Table 11: Loss-based membership inference against the downstream XGBoost learner trained on the real records (Real), on the TARDIS synthetic dataset, and on the TabDiff synthetic dataset. Lower is better; 0.5 is chance level for the attack AUC.

	Attack AUC ↓			TPR@1%FPR ↓	
Dataset	Real	TARDIS	TabDiff	Real	TARDIS
Solar Flare	0.552	0.502	0.449	0.030	0.047
Contraceptive	0.821	0.553	0.526	0.011	0.000
Diabetic Retin.	0.855	0.610	0.548	0.058	0.035
Infrared Therm.	0.997	0.498	0.494	0.967	0.014
Abalone	0.969	0.481	0.493	0.142	0.007
Magic	0.592	0.499	0.504	0.000	0.000
Student Perf.	0.779	0.713	0.646	0.028	0.000
Insurance	0.967	0.543	0.543	0.384	0.002
Adult	0.562	0.504	0.507	0.000	0.000
Default	0.676	0.537	0.524	0.009	0.007
Beijing	0.838	0.544	0.500	0.064	0.013
Diamonds	0.648	0.537	0.511	0.000	0.009
Music	0.918	0.600	0.537	0.059	0.019
Bank	0.534	0.521	0.505	0.000	0.000
News	0.764	0.491	0.505	0.035	0.010
Mean	0.765	0.542	0.520	0.119	0.011

Table 12: Fraction of synthetic samples that exactly match a training record, and mean nearest-neighbor distance to the training records in z-scored one-hot space.

	Exact-match fraction ↓		Mean NN distance ↑	
Dataset	TARDIS	TabDiff	TARDIS	TabDiff
Solar Flare	0.6842	0.7018	0.495	0.573
Contraceptive	0.0215	0.0204	0.757	0.801
Diabetic Retin.	0.0000	0.0000	2.209	1.745
Infrared Therm.	0.0000	0.0000	3.877	4.202
Abalone	0.0000	0.0000	0.409	0.421
Magic	0.0000	0.0000	0.656	0.660
Student Perf.	0.4478	0.4591	2.305	2.243
Insurance	0.0000	0.0000	1.102	0.934
Adult	0.0000	0.0000	1.658	1.673
Default	0.0010	0.0011	1.023	1.025
Beijing	0.0000	0.0000	0.585	0.586
Diamonds	0.0000	0.0000	0.348	0.350
Music	0.0000	0.0000	1.163	1.160
Bank	0.0000	0.0000	1.749	1.412
News	0.0000	0.0000	4.193	3.558
Mean	0.0770	0.0788	1.502	1.423

Fine-tuning and search. A fine-tuning pass over 10% of the training schedule would already cost 13 to 35 minutes per dataset on the L40S, comparable to or above the entire refinement, and would yield a distinct model per configuration; TARDIS reuses one frozen checkpoint and requires no parameter updates at apply time. The configuration search is a one-time cost that can be reduced: the sealed protocol reaches positive transfer with a median of 50 trials (Appendix K), and the fixed Stage III configuration of Appendix J.3 requires no Stage III search.

I Cardinality and Selection Controls

The controls separate the contributions of the candidate pool, the selection criterion, and the output cardinality. They use the datasets, splits, backbone checkpoints, and evaluation protocol of Section 4

Table 13: Per-dataset TARDIS runtime breakdown. Stage II is the generation of D_{cand} (guided where enabled) ($M = 50$); Stage III is the post-hoc selection (Chamfer Sampler or MDSampler); GKD is the soft-label distillation overhead, applicable only when the configuration enables it. Datasets sorted by total runtime ascending.

Dataset	Stage II (min)	Stage III (min)	GKD (min)	Total (min)
Solar Flare	0.7	0.1	0.1	0.9
Insurance	0.8	0.2	–	1.1
Student Perf.	1.0	0.2	–	1.4
Diabetic Retin.	1.2	0.3	–	1.7
Contraceptive	1.8	0.1	–	2.0
Infrared Therm.	1.3	0.6	–	2.1
Abalone	2.5	1.0	–	3.8
Magic	9.8	4.9	–	16.1
Music	14.3	11.4	–	29.8
Adult	22.9	7.0	2.3	32.2
Default	38.5	7.1	–	47.9
Beijing	45.3	4.9	–	53.7
Diamonds	43.3	16.0	–	65.3
News	36.3	29.8	–	77.7
Bank	65.7	10.1	–	79.4

Table 14: Apply-time cost on the GTX 1080 Ti (Table 13) and in the rerun on an NVIDIA L40S, TabDiff training time on the L40S, and end-to-end TARDIS run on the L40S (pool generation, soft labelling, Stage III samplers with scoring, final model training).

	Apply (min)			TabDiff training	Apply /	End-to-end
Dataset	1080 Ti	L40S	Speedup	L40S (h)	training	L40S (min)
Solar Flare	0.9	0.2	5.4×	2.2	0.1%	0.5
Insurance	1.1	0.7	1.7×	2.2	0.5%	1.1
Student Perf.	1.4	0.7	2.0×	2.2	0.5%	2.0
Diabetic Retin.	1.7	0.5	3.5×	2.2	0.4%	1.4
Contraceptive	2.0	0.6	3.5×	2.9	0.3%	1.5
Infrared Therm.	2.1	0.6	3.7×	2.2	0.4%	1.4
Abalone	3.8	0.6	6.0×	2.7	0.4%	1.6
Magic	16.1	3.6	4.5×	3.4	1.8%	9.9
Music	29.8	10.3	2.9×	4.5	3.8%	40.8
Adult	32.2	6.2	5.2×	5.8	1.8%	8.8
Default	47.9	6.7	7.1×	5.3	2.1%	13.2
Beijing	53.7	6.4	8.3×	4.9	2.2%	13.7
Diamonds	65.3	17.9	3.6×	4.4	6.8%	62.3
News	77.7	15.6	5.0×	4.5	5.8%	36.4
Bank	79.4	8.2	9.7×	5.6	2.4%	18.9
Median	16.1	3.6	4.5×	3.4	1.8%	8.8

and run on NVIDIA L40S GPUs; the Real, TabDiff, TARDIS$_{K=1}$, and TARDIS columns are those of Table 1.

I.1 Controls at Matched Cardinality

Table 15 compares TARDIS against four controls that produce $K \cdot N_r$ rows with the K selected per dataset, and against Guidance-only, which produces N_r rows. *Real-bootstrap* resamples D_{real} with replacement to the same cardinality, isolating the effect of training-set size. *TabDiff oversampled* draws $K \cdot N_r$ rows from the unguided backbone and keeps them without selection, isolating the effect of generating more synthetic data. *Chamfer without Stage II* applies the Stage III Chamfer Sampler to an unguided candidate pool, isolating the discrete refinement level. *Random from pool* draws uniformly from the same candidate pool that TARDIS ranks, isolating the selection criterion from the quality of the pool. *Guidance-only* sets $M = K = 1$, so no selection takes place; it is defined only on the nine datasets whose selected configuration enables Stage II guidance.

Real-bootstrap remains far from TARDIS on the datasets with the largest gains (Diabetic Retinopathy, Student Performance), so training-set size does not explain them. Random selection from the same pool underperforms TARDIS on 14 of 15 datasets. Chamfer selection without Stage II recovers a large share of the gain on several datasets; on Infrared Thermography and Insurance neither refinement level alone approaches the full configuration.

Table 15: Controls at the final cardinality $K \cdot N_r$ of each selected configuration. AUROC and weighted-AUROC ↑ for classification, RMSE ↓ for regression. *Guid. only* sets $M = K = 1$ and is defined only where the selected configuration enables Stage II guidance; dashes mark the six no-guidance datasets of Table 8. Best per row in bold.

Dataset	Real	Real-boot.	TabDiff over.	Chamfer w/o II	Rand. pool	Guid. only	TARDIS
Regression — RMSE ↓							
Solar Flare	0.1025	0.0960	0.0843	0.0742	0.1041	–	**0.0620**
Infrared Therm.	0.0541	0.0534	0.0660	0.0638	0.0517	0.0615	**0.0482**
Abalone	0.0810	0.0787	0.0780	0.0799	0.0778	–	**0.0740**
Insurance	0.1180	0.1351	0.1563	0.1536	0.1276	0.1445	**0.1100**
Beijing	**0.0330**	0.0342	0.0388	0.0390	0.0379	0.0498	0.0370
News	0.0115	0.0121	0.0099	0.0098	0.0109	0.0119	**0.0096**
Binary classification — AUROC ↑							
Diabetic Retin.	0.6976	0.7185	0.7718	0.8300	0.7475	0.6797	**0.9037**
Adult	**0.9270**	0.9210	0.9104	0.9102	0.9117	–	0.9240
Default	0.7640	0.7687	0.7687	0.7723	0.7712	0.7630	**0.7750**
Magic	0.9560	0.9481	0.9589	0.9571	0.9573	–	**0.9650**
Bank	0.9290	0.9289	0.9297	**0.9335**	0.9315	–	0.9331
Multiclass classification — weighted-AUROC ↑							
Student Perf.	0.7392	0.7309	0.8212	0.8068	0.7279	0.7164	**0.8939**
Contraceptive	0.6491	0.6646	0.6797	0.6602	0.6932	0.6711	**0.7077**
Music	0.7290	0.7296	0.7357	0.7363	0.7333	–	**0.8400**
Diamonds	0.9430	0.9398	0.9420	**0.9435**	0.9434	0.9326	0.9430

I.2 Controls at Matched Output Size

In Table 16 every arm emits N_r rows; TARDIS$_{K=1}$ keeps the $M = 50$ pool and the Stage III selection.

At N_r rows, TARDIS$_{K=1}$ improves over TabDiff on 11 of 15 datasets but not significantly over real data (Section 4.1), and Guidance-only improves over TabDiff on three of its nine datasets. Exceeding real-data utility therefore requires the larger output cardinality, which neither real-data resampling nor oversampling without selection exploits on its own (Section I.1).

J Deconfounding Pool Size, Guidance, and Search

The analyses use the evaluation protocol of Section 4 on eight corpora spanning the three task types (Sections J.1 and J.2) and on all fifteen (Section J.3). Values are means over 4 seeds on NVIDIA L40S GPUs, except those taken from Table 1.

J.1 Pool Size

At $K = 1$ a larger pool yields no consistent benefit: the best M varies by dataset, with small margins on most corpora (Table 17). At $K = 10$, inside the range selected by the search ($M \geq K$; Table 18), the spread across M is of the order of the seed-to-seed standard deviation (ratio: median 0.49, range $[0.33, 1.27]$); only on Diabetic Retinopathy is a larger pool worse by more than the seed noise. The best configuration at $K = 10$ improves over the best at $K = 1$ on 6 of 8 corpora.

Table 16: Controls at matched output size N_r. AUROC and weighted-AUROC ↑ for classification, RMSE ↓ for regression. *Guidance-only* sets $M = K = 1$ and is defined only where the selected configuration enables Stage II guidance. Best per row in bold.

Dataset	Real	TabDiff	Guidance-only (M=K=1)	TARDIS$_{K=1}$ (M=50, K=1)
Regression — RMSE ↓				
Solar Flare	0.1025	0.0923	–	**0.0869**
Infrared Therm.	**0.0541**	0.0585	0.0615	0.0562
Abalone	0.0810	0.0790	–	**0.0750**
Insurance	**0.1180**	0.1320	0.1445	0.1430
Beijing	**0.0330**	0.0510	0.0498	0.0510
News	0.0115	0.0119	0.0119	**0.0100**
Binary classification — AUROC ↑				
Diabetic Retin.	**0.6976**	0.6671	0.6797	0.6697
Adult	**0.9270**	0.9130	–	0.9220
Default	0.7640	0.7650	0.7630	**0.7680**
Magic	**0.9560**	0.9400	–	0.9490
Bank	**0.9290**	0.8560	–	0.8630
Multiclass classification — weighted-AUROC ↑				
Student Perf.	0.7392	**0.7453**	0.7164	0.6796
Contraceptive	0.6491	**0.6838**	0.6711	0.6770
Music	0.7290	0.8230	–	**0.8330**
Diamonds	**0.9430**	0.9300	0.9326	0.9340

Table 17: Sweep of the pool multiplier M at output size $K = 1$.

Dataset	Metric	$M = 5$	$M = 10$	$M = 25$	$M = 50$
Solar Flare	RMSE	0.0995	0.0942	0.0962	0.0869
Infrared Therm.	RMSE	0.0602	0.0636	0.0607	0.0562
Diabetic Retin.	AUROC	0.6319	0.6421	0.5769	0.6697
Student Perf.	AUROC	0.7095	0.7154	0.7497	0.6796
Contraceptive	AUROC	0.6710	0.6575	0.6808	0.6770
Insurance	RMSE	0.1382	0.1349	0.1371	0.1361
Abalone	RMSE	0.0820	0.0819	0.0847	0.0802
Magic	AUROC	0.9421	0.9420	0.9400	0.9401

J.2 One Component at a Time

Table 19 changes one component at a time. On the three configurations without guidance both pools are unguided, so their rung-1 to rung-2 difference (0.8% to 4.3%) is a noise band for pool generation.

On the guided datasets, Δ guidance ranges from -3.1% to $+3.2\%$ (median -0.6%), within the noise band, whereas Δ Stage-III search is positive on all eight corpora (median $+8.9\%$) and exceeds the band on five; within rung 3, sampler and cardinality vary together.

Together with Section J.1 and Appendix I, these results attribute the benefit to cardinality and selection; guidance is enabled per dataset by the search, and M yields no consistent benefit.

J.3 Per-Dataset Search against a Fixed Global Configuration

The archived per-dataset pools retain the searched Stage II guidance where enabled (unguided on Insurance). The per-dataset search wins on 14 of 15 datasets (Table 20), with the largest margins on Diabetic Retinopathy and Student Performance.

J.4 Practical Defaults

Under a budget of $K \approx 1$, $M \in [5, 10]$ suffices (Table 17), and the search selects GKD only at $K \leq 1$ (Table 8). All results in the main text use $M = 50$.

Table 18: Sweep of the pool multiplier M at output size $K = 10$, with the spread across M and the seed-to-seed standard deviation.

Dataset	Metric	$M = 10$	$M = 25$	$M = 50$	Spread across M	Seed std
Solar Flare	RMSE	0.0920	0.0975	0.0945	0.0055	0.0125
Infrared Therm.	RMSE	0.0523	0.0506	0.0529	0.0023	0.0045
Diabetic Retin.	AUROC	0.7673	0.7443	0.7128	0.0545	0.0429
Student Perf.	AUROC	0.7291	0.7179	0.7318	0.0139	0.0136
Contraceptive	AUROC	0.6936	0.6772	0.7031	0.0259	0.0264
Insurance	RMSE	0.1279	0.1202	0.1207	0.0076	0.0166
Abalone	RMSE	0.0775	0.0776	0.0767	0.0009	0.0027
Magic	AUROC	0.9524	0.9535	0.9546	0.0022	0.0064

Table 19: Deconfounding ladder at fixed Chamfer selection. Rung 1: unguided pool, $K = 10$. Rung 2: per-dataset pool, $K = 10$. Rung 3: full configuration of Table 1. *Configurations without guidance: both pool arms are unguided, and their difference is a noise band. †The archived Insurance pool does not include its searched guidance, so rung 2 uses a pool generated with the selected configuration.

Dataset	1. Unguided, K=10	2. Per-dataset pool, K=10	3. TARDIS	Δ guidance	Δ Stage-III search
Solar Flare*	0.0959	0.0945	0.0620	–	$+34.4\%$
Infrared Therm.	0.0526	0.0529	0.0482	-0.6%	$+8.9\%$
Diabetic Retin.	0.7355	0.7128	0.9037	-3.1%	$+26.8\%$
Student Perf.	0.7178	0.7318	0.8939	$+1.9\%$	$+22.2\%$
Contraceptive	0.6811	0.7031	0.7077	$+3.2\%$	$+0.7\%$
Insurance	0.1172	0.1207†	0.1100	-3.0%	$+8.8\%$
Abalone*	0.0801	0.0767	0.0740	–	$+3.5\%$
Magic*	0.9620	0.9546	0.9650	–	$+1.1\%$

K Sealed Evaluation Protocol

In the main protocol, the validation split serves both XGBoost early stopping and the configuration search (Section 4). On the six smallest corpora, the sealed protocol sets aside a 5% holdout before any processing, which the search never sees. For each of four resamples per corpus (24 runs), the TabDiff backbone is retrained from scratch and the search runs on its own splits with a reduced budget (median 50 trials, against 500 or $1{,}000$ in the main protocol); Real, TabDiff, and TARDIS are each evaluated once on the holdout, on NVIDIA L40S GPUs.

Over the 24 runs (Table 21), the median $\Delta\%$ is $+4.9$ against Real (positive on 18) and $+7.1$ against TabDiff (positive on 22). With holdouts of 30 to 75 rows, single runs are noisy (-29.9% to $+16.3\%$ against Real).

L Transfer to Other Downstream Learners

On the TARDIS synthetic datasets of ten corpora, searched with XGBoost as validation utility, we train three other learners without re-tuning: CatBoost [29], an MLP with hidden layers $(128, 64)$, and a linear model (logistic regression for classification, ridge regression for regression). The study uses one data split of the protocol of Section 4; CatBoost and MLP entries are medians over 5 initialization seeds, and the linear model is fit once, on NVIDIA L40S GPUs.

The MLP improves over its real-data counterpart on all ten corpora (median $+2.1\%$), whereas CatBoost ($+0.2\%$) and the linear models (-0.1%) remain at parity (Table 22); for the linear models this is consistent with their insensitivity to the local manifold structure that the refinement restores.

M Worked Example

This appendix follows Diabetic Retinopathy through the pipeline of Figure 1, with its configuration from Table 8.

Table 20: Fixed global configuration (Chamfer Sampler, $K = 10$, no GKD) on the archived per-dataset pools against the per-dataset search of Table 1. Best per row in bold.

Dataset	Metric	Fixed global configuration	Per-dataset search
Solar Flare	RMSE	0.0945	**0.0620**
Infrared Therm.	RMSE	0.0529	**0.0482**
Abalone	RMSE	0.0767	**0.0740**
Insurance	RMSE	0.1377	**0.1100**
Beijing	RMSE	0.0398	**0.0370**
News	RMSE	0.0109	**0.0096**
Diabetic Retin.	AUROC	0.7128	**0.9037**
Adult	AUROC	0.9160	**0.9240**
Default	AUROC	**0.7777**	0.7750
Magic	AUROC	0.9546	**0.9650**
Bank	AUROC	0.9297	**0.9331**
Student Perf.	AUROC	0.7318	**0.8939**
Contraceptive	AUROC	0.7031	**0.7077**
Music	AUROC	0.7337	**0.8400**
Diamonds	AUROC	0.9422	**0.9430**

Table 21: Sealed protocol on the six smallest corpora, four resamples each. Holdout utility (weighted-AUROC for Contraceptive and Student Performance, AUROC for Diabetic Retinopathy, RMSE for the regression corpora) and sign-corrected $\Delta\%$ against Real and TabDiff.

Dataset	Split	Real	TabDiff	TARDIS	$\Delta\%$ vs. Real	$\Delta\%$ vs. TabDiff
Contraceptive	0	0.7602	0.7420	0.7873	$+3.6$	$+6.1$
Contraceptive	1	0.6801	0.6727	0.6818	$+0.3$	$+1.4$
Contraceptive	2	0.6591	0.6779	0.6930	$+5.1$	$+2.2$
Contraceptive	3	0.7268	0.7475	0.7469	$+2.8$	-0.1
Diabetic Retin.	0	0.7851	0.6995	0.8327	$+6.1$	$+19.0$
Diabetic Retin.	1	0.7913	0.7727	0.7793	-1.5	$+0.9$
Diabetic Retin.	2	0.7847	0.6083	0.7333	-6.5	$+20.5$
Diabetic Retin.	3	0.7097	0.6900	0.7742	$+9.1$	$+12.2$
Infrared Therm.	0	0.0479	0.0668	0.0540	-12.8	$+19.2$
Infrared Therm.	1	0.0516	0.0817	0.0493	$+4.6$	$+39.7$
Infrared Therm.	2	0.0478	0.0667	0.0621	-29.9	$+7.0$
Infrared Therm.	3	0.0496	0.0594	0.0478	$+3.6$	$+19.5$
Insurance	0	0.1498	0.1495	0.1389	$+7.3$	$+7.1$
Insurance	1	0.1413	0.1233	0.1201	$+15.0$	$+2.6$
Insurance	2	0.1570	0.2168	0.1855	-18.2	$+14.4$
Insurance	3	0.1141	0.1362	0.1051	$+7.9$	$+22.8$
Solar Flare	0	0.1236	0.1246	0.1143	$+7.5$	$+8.2$
Solar Flare	1	0.1034	0.0981	0.0865	$+16.3$	$+11.8$
Solar Flare	2	0.1010	0.0908	0.0873	$+13.6$	$+3.8$
Solar Flare	3	0.1022	0.1052	0.1027	-0.5	$+2.4$
Student Perf.	0	0.8058	0.8168	0.8179	$+1.5$	$+0.1$
Student Perf.	1	0.6912	0.7308	0.7821	$+13.2$	$+7.0$
Student Perf.	2	0.6808	0.7501	0.7293	$+7.1$	-2.8
Student Perf.	3	0.7038	0.7023	0.7950	$+13.0$	$+13.2$
Median over the 24 runs					$+4.9$	$+7.1$

Stage I. The dataset has 1,151 rows; the 90% training split yields $N_r = 1{,}035$ rows, and Stage I draws $M \cdot N_r = 51{,}750$ noise vectors from the latent prior of TabDiff.

Stage II. Reverse diffusion denoises the pool under the score perturbation of Equation 4, with φ the encoder of a β-VAE of latent dimension $d = 4$ trained on D_{real}, the cosine family for $\Gamma(t)$, and class-conditional reference batches with batching ratio $1/2$.

Stage III. The Chamfer Sampler ranks batches of the candidate pool D_{cand} by $\mathcal{C}(B_k, D_{\text{real}})$ in the same φ-space and retains the top-ranked batches until $K \cdot N_r \approx 23{,}290$ rows are collected ($K = 22.5$); GKD is disabled.

Table 22: Zero-shot transfer to other learners: sign-corrected $\Delta\%$ of each learner trained on the TARDIS synthetic dataset with respect to the same learner trained on the real data.

Dataset	CatBoost	MLP (128, 64)	Linear (LogReg / Ridge)
Contraceptive	-0.8%	$+0.9\%$	$+0.2\%$
Diabetic Retin.	$+5.8\%$	$+9.4\%$	-4.8%
Infrared Therm.	$+2.6\%$	$+34.3\%$	-2.7%
Abalone	$+2.4\%$	$+10.7\%$	$+0.6\%$
Magic	-0.1%	$+1.5\%$	-0.1%
Student Perf.	$+0.2\%$	$+1.6\%$	$+0.9\%$
Default	$+0.0\%$	$+1.4\%$	$+0.6\%$
Diamonds	$+0.2\%$	$+2.0\%$	-0.8%
Music	$+0.6\%$	$+2.2\%$	-0.0%
News	-0.3%	$+11.2\%$	-0.9%
Median	$+0.2\%$	$+2.1\%$	-0.1%
TARDIS better on	7/10	10/10	4/10

Outcome. XGBoost trained on these rows reaches a test AUROC of 0.9037, against 0.6976 when trained on the real rows (Table 1).

N Broader Impact

A primary motivation for TARDIS is to enable downstream training in settings where direct access to real tabular records is restricted by privacy regulation, contractual constraints, or institutional barriers. Synthetic tabular generation supports such use cases, but the synthetic–real utility gap has historically forced practitioners to choose between privacy-preserving access and faithful downstream performance. TARDIS closes this gap at inference time without modifying a pre-trained backbone, reducing the friction between privacy-driven data sharing and downstream learner quality. The framework inherits TabDiff's handling of mixed-type features and class structure; the proximity proxies we report (Distance to Closest Record, Nearest-Neighbor Distance Ratio) and the near-duplicate counts stay at the level of the underlying backbone (Section 4.3, Appendix G.3).

We acknowledge that synthetic data inherits the biases of the corpus on which the backbone is trained, and that the bidirectional Chamfer functional, which steers candidates toward the support of the real-data manifold, can in principle reinforce existing demographic, class, or representational imbalances rather than mitigate them. The two-regime structure surfaced by our search (high-K utility maximization, low-K utility preservation under cardinality constraints) exposes explicit hyperparameters to control the synthetic distribution's relationship to the real one, but does not constitute a fairness guarantee. We make no formal differential-privacy claim; the empirical privacy proxies we report capture distance-to-training-record signals but do not bound information leakage in the cryptographic sense. Practitioners deploying TARDIS in privacy-regulated or fairness-sensitive settings should treat these metrics as necessary rather than sufficient, complement them with task-specific fairness audits and membership-inference evaluations, and consider the differentially-private variant outlined in Section 5 when formal guarantees are required. The present study operates on standard public benchmarks at a limited data scale and is oriented toward methodological analysis; we do not identify pathways to direct misuse, but recognize that scaling inference-time refinement of generative tabular models warrants ongoing scrutiny.

NeurIPS Paper Checklist

1. **Claims**

   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

   Answer: [Yes]

   Justification: The abstract and introduction present TARDIS as an inference-time search framework over a frozen pre-trained backbone. Each claim is supported in Section 4: utility above real data at the cardinality selected by the search (median $+8.6\%$, mean $+9.8\%$, 95% CI of the mean $[+3.8, +16.4]$, Wilcoxon $p = 0.008$) and above TabDiff on all 15 datasets (Section 4.1); BCR-aligned configurations selected on 9 of 15 datasets (Section 4.2); fidelity, diversity, and privacy proxies at the level of the backbone (Section 4.3). The abstract also states that at output size matched to the real data TARDIS reaches parity with real data (Appendix I).

2. **Limitations**

   Question: Does the paper discuss the limitations of the work performed by the authors?

   Answer: [Yes]

   Justification: Section 5 states that the full pipeline is validated on a single backbone (TabDiff); that Stage II is untested on TabSyn and TabDDPM and does not apply to autoregressive or variational decoders, while Stage III alone yields no consistent gain on CTAB-GAN+ and TVAE pools; that configurations are searched with XGBoost and other learners are evaluated zero-shot on ten corpora; that M is fixed at 50 and swept only at $K \in \{1, 10\}$ on eight corpora; that run-to-run variance reflects pool generation only; and that privacy is assessed through proximity proxies and a membership-inference attack, without formal differential-privacy guarantees.

3. **Theory assumptions and proofs**

   Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

   Answer: [N/A]

   Justification: The paper does not include formal theorems. Appendix A provides the structural properties of the symmetric Chamfer functional $\mathcal{C}$ (non-negativity, identity of indiscernibles, symmetry) with brief inline justifications, but it is not intended as a theorem.

4. **Experimental result reproducibility**

   Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

   Answer: [Yes]

   Justification: The TPE search space is fully specified in Table 7, with per-dataset selected configurations in Table 8. Stage II schedules are defined with closed-form formulas (Appendix C); Stage III samplers in Appendix D; GKD details in Appendix E. Data splits, preprocessing, and downstream learner configuration are documented in Section 4; per-dataset structural attributes in Table 6.

5. **Open access to data and code**

   Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

   Answer: [Yes]

   Justification: Code, Optuna search spaces, and β-VAE/sampler implementations are available at `https://github.com/[TODO]`. The 15 benchmarks are public and cited in Table 6; the search space, schedule families, sampler definitions, and per-dataset configurations are reported in Appendices C, D, and F.

6. **Experimental setting/details**

Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

Answer: [Yes]

Justification: Data splits, downstream learner (XGBoost), TSTR protocol, and aggregation methodology ($\Delta\%$ with bootstrap CI and Wilcoxon paired tests) are documented in Section 4. The TPE search space and selection procedure are reported in Appendix F; backbone selection rationale in Appendix B.

7. **Experiment statistical significance**

Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

Answer: [Yes]

Justification: Aggregate improvements are reported with bootstrap 95% confidence intervals of the mean and Wilcoxon signed-rank paired tests across the 15 datasets ($p = 0.008$ vs. Real, $p < 10^{-4}$ vs. TabDiff; Table 1). Each configuration is run 5 times; the downstream learner is deterministic, so run-to-run variance reflects pool generation only (Section 4). A sealed protocol with a holdout never seen by the search is reported in Appendix K.

8. **Experiments compute resources**

Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

Answer: [Yes]

Justification: Unless stated otherwise, experiments run on a single Nvidia GTX 1080 Ti (12 GB; Section 4), where applying a selected configuration takes 0.9 to 79.4 minutes per dataset (median 16.1; Table 13). The analyses of Appendices B (transfer to other generators), G (membership inference), H (rerun), and I–L run on NVIDIA L40S GPUs, where training the TabDiff backbone takes 2.2 to 5.8 hours per dataset (Table 14).

9. **Code of ethics**

Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics?

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Justification: The paper builds on publicly available tabular benchmarks (UCI, Kaggle, PhysioNet) and a publicly released backbone (TabDiff). No human subjects, crowdsourcing, or sensitive data collection is involved. Privacy implications of synthetic tabular generation are addressed empirically in Section 4.3 and discussed in Appendix N.

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Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

Answer: [Yes]

Justification: Appendix N discusses positive impacts (enabling downstream training in privacy-restricted settings without retraining the backbone, with proximity proxies and near-duplicate counts at the level of the backbone) and potential concerns (biases inherited from the training corpus, absence of formal privacy guarantees).

11. **Safeguards**

Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse?

Answer: [N/A]

Justification: TARDIS is a refinement framework operating on top of an existing tabular diffusion backbone; it does not release a new generative model, language model, or scraped dataset. The TabDiff backbone is publicly released by its original authors and is not redistributed by this work.

12. **Licenses for existing assets**

Question: Are the creators or original owners of assets used in the paper properly credited and are the license and terms of use explicitly mentioned and properly respected?

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Justification: The 15 benchmarks are public datasets from the UCI Machine Learning Repository, Kaggle, and PhysioNet, cited in Table 6; the UCI datasets are distributed under CC BY 4.0, and the Kaggle and PhysioNet datasets are used under the terms stated on their pages. The TabDiff [31] backbone, comparison baselines TabSyn [43] and TabularARGN [33], downstream learner XGBoost [10], and evaluation metrics are cited with their original publications. The TPE optimizer follows [4] and the cyclical KL annealing schedule follows [15].

13. **New assets**

Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

Answer: [Yes]

Justification: The TARDIS framework is fully documented in the appendix: BCR principle (Appendix A), TPE search space (Appendix F), Stage II schedules (Appendix C), Stage III samplers (Appendix D), and GKD (Appendix E).

14. **Crowdsourcing and research with human subjects**

Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

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Answer: [N/A]

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16. **Declaration of LLM usage**

Question: Does the paper describe the usage of LLMs if it is an important, original, or non-standard component of the core methods in this research?

Answer: [N/A]

Justification: LLMs are not components of the TARDIS framework.